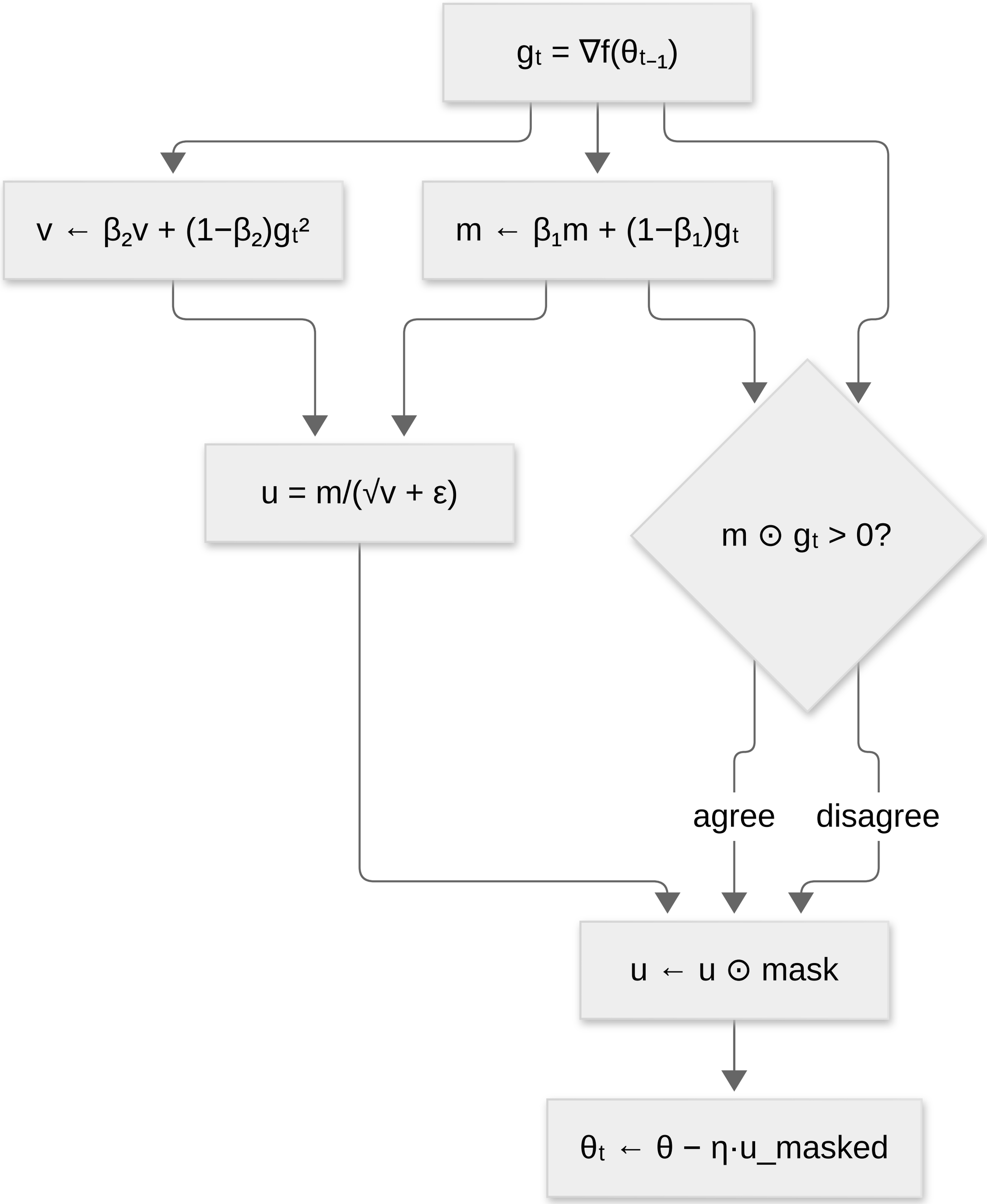

$g_t = \nabla f(\theta_{t-1})$
$v \leftarrow \beta_2 v + (1-\beta_2) g_t^2$
$m \leftarrow \beta_1 m + (1-\beta_1) g_t$
$u = m/(\sqrt{v} + \varepsilon)$
$m \odot g_t > 0$?
agree
disagree
$u \leftarrow u \odot \text{mask}$
$\theta_t \leftarrow \theta - \eta \cdot u_masked$